

TECHNICAL & COMMERCIAL PROPOSAL

Design, Supply, Installation, Testing,
Commissioning & Maintenance of Cold Rooms

3 Cold Rooms 1,2,3 (+0C → +5C) + Corridor +10C

Submitted To:

Salam Petroleum Services

Attn: Mr. Yousef Saleh

Proposal Reference: MES/QTN/SP-CR-2026/140 Rev.2

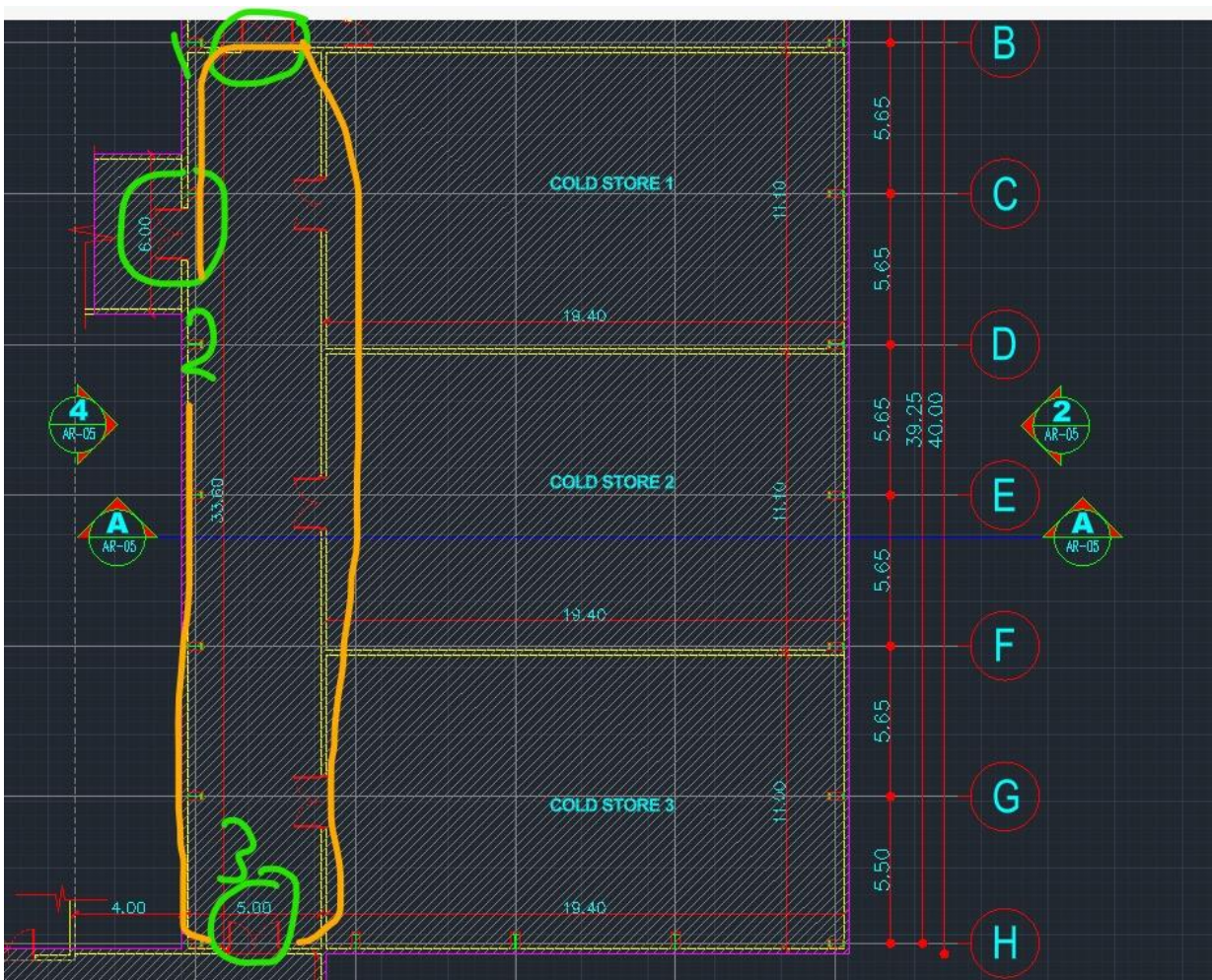
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1. EXECUTIVE SUMMARY

Metri Engineering Services W.L.L. (MES) is pleased to submit this technical and commercial proposal (Revision 2- (3 Cold Rooms Chiller application + Corridor) for the design, supply, installation, testing, commissioning, and maintenance of cold storage rooms at the Salam Petroleum Services facility in Doha, Qatar.

The facility comprises (3) individually controlled cold rooms (Chiller range temperature +0C→+5C) + Corridor (+10C) 33.5mx 5m x 7m height.

Our proposal incorporates Condensing Units with Bitzer (Germany) compressor technology as specified, Evaporators made in Europe LUVE, cold store 100mm thick PUR sandwich panel construction with PPGI 0.5mm made in Qatar.



2. SCOPE OF WORK

2.1 Included in Scope

1. Complete engineering design including heat load calculations, equipment selection, refrigerant piping, and control requirements
2. Supply and installation of **PUR** insulated sandwich panels for all walls and ceilings (100mm) made in Qatar.
3. Supply and installation of **XPS** floor insulation boards made in Qatar (50mm, 2 layers as per temperature zone). Concrete ontop and epoxy by Client.
4. Supply and installation of manual sliding cold room doors (2.0m W × 3.0m H) for all three cold rooms and corridor. Total 6 nos Doors
5. Supply, installation, and commissioning of **Condensing Units with Bitzer (Germany)** compressor-based refrigeration systems including condensing units, with matching **evaporators made in Europe**, expansion valves, controls, and interconnecting refrigerant piping.
6. Complete lighting (LED cold-rated), control panels, and safety systems
7. Testing & commissioning of all systems.
8. As-built drawings, O&M manuals, and operator training.
9. **Warranty:**
 - Sandwich panels 10 years.
 - Refrigeration system: 1 year for compressors, evaporators, refrigeration pipes and accessories. Control panels and installation of electrical contactors.
 - AMC after warranty period will be charged 3,000 Q.r Monthly excluding any parts or materials

2.2 Exclusions

- Civil works and building structure and any steel supports
- Epoxy flooring
- Concrete works (slab, foundations, screed)
- Painting
- Building permits and approvals (by client)
- Electrical supply up to the main isolator
- Fire suppression systems
- Racking/shelving inside cold rooms

3. DESIGN BASIS & HEAT LOAD SUMMARY

3.1 Design Conditions

Parameter	Value	
Outside Design DB Temperature	46°C	
Outside Design WB Temperature	28°C	
Ground Temperature	35°C	
Equipment Operating Hours	22 hrs/day	2 hrs defrost
Cold Room 1,2,3	+0°C → +5°C	
Corridor	+10°C → +12°C	

3.3 Room Schedule

Room	Temp (°C)	Dims (m)	Panel	Qty	Application
Cold Store 1,2,3	+0°C → +5°C	19.4×11.1×7	100mm	3	Chilled storage
Corridor	+10°C → +12°C	33.6x5x7	100mm	1	Chilled Corridor

3.4 Heat Load Calculation Summary

Room	Trans. (kW)	Product (kW)	Infiltration (kW)	Internal (kW)	Total/Room (kW)
Cold Store 1,2,3 (0°C)	17.49	21.92	4.30	3.22	51.22
Corridor +10°C → +12°C					28

4. TECHNICAL SPECIFICATIONS

4.1 Sandwich Panel Insulation

All cold room enclosures shall be factory-manufactured **PUR** insulated sandwich panels with pre-painted galvanized steel facings (0.5mm thickness), tongue-groove jointing, and silicone sealing:

Zone	Thickness	U-value (W/m²K)	Application
Chiller (0°C)	100mm	0.2128	walls/ceiling
Corridor +10°C → +12°C	100mm	0.2128	walls/ceiling

4.2 Floor Insulation

Zone	Layers	Total Area	U-value (W/m²K)
Chiller Rooms & Corridor (0°C)	2 × 50mm	1,300 m² (incl. layers)	0.3030
Corridor +10°C → +12°C	2 × 50mm	336m² (incl. layers)	0.3030

4.3 Doors

Item	Specification
Cold Room Doors (3 nos.)	Manual sliding, 2.0m W × 3.0m H, PUR insulated matching room panel thickness, heated frames on Chiller doors
Corridor Doors (3 nos)	
Hardware & Safety	SS handles, interior safety release
Accessories	PVC strip curtains on all 3 doors

4.4 Refrigeration Systems

Room	Compressor	Evap Temp	Cooling Capacity/room	Refrigerant	Defrost	Qty
Cold St 1,2,3 (0°C)	Bitzer Semi-Hermetic	-7°C	59 kW ea	R-404A	Electric	2 nos of CU each room + 2 nos of matching Evaporators each room
Corridor +10°C → +12°C	Bitzer Semi- Hermetic	+0°C	30Kw	R-404A	Electric	2 nos of CU + 2 nos of matching Evaporators

4.5 Evaporators, Controls & Electrical

Ceiling-mounted industrial unit coolers, Control panel for each room. Each room receives a controller. Electrical limited to include distribution after the isolator, LED cold-rated lighting (IP65)

5. EQUIPMENT SCHEDULE

Item	Description	For Room	Cooling Capacity each Room	Origin	Proposed Qty of equipment per room	Total QTY
1	Bitzer Semi-Hermetic Condensing Unit	Cold St 1,2,3 +0C	59 kW	BITZER GERMANY	2 (each cooling capacity 29.5kw)	6
2	Evaporator – med temp	Cold St 1,2,3 +0C	Matching with above CU	LUVE EUROPE	2	6
3	Bitzer Semi-Hermetic Condensing Unit	Corridor +10°C → +12°C	30 kW	BITZER GERMANY	2 (each cooling capacity 15kw)	2
4	Evaporator – med temp	Corridor +10°C → +12°C	Matching with above CU	LUVE EUROPE	2	2
5	Cold Room Sliding Door 2.0×3.0m	All rooms+Corridors	-		1	6
6	Control Panels	All	-		1	4

6. BILL OF QUANTITIES & PRICING

Item	Description	Unit	Qty	Rate (QAR)	Amount (QAR)
A	INSULATED PANEL WORKS				
A1	100mm PUR Sandwich Panels – Corridor & Cold Stores (walls + ceiling)	m ²	2,250	155	348,750
A2	XPS Floor Insulation 2×50mm – Chiller rooms (total board area)	m ²	1,680	50	84,000
A3	Panel accessories: flashings, trims, corner pieces, sealant, vapour barrier	Lot	1	-	15,000
	SUB-TOTAL A – INSULATED PANEL WORKS				<u>447,750</u>
B	COLD ROOM DOORS				
B1	Manual Sliding Door 2.0×3.0m – 100mm insulated (Cold Stores)	No.	6	12,000	72,000
B2	PVC Strip Curtains (all doors)	No.	6	2,000	12,000
	SUB-TOTAL B – DOORS				<u>84,000</u>
C	REFRIGERATION SYSTEMS				
C1	Bitzer Semi-Hermetic Condensing Unit – Cold Stores (-7°C evap) 29.5kw	No.	6	95,000	570,000
C2	Unit Coolers – Medium temp (Cold Stores) matching with above CU	No.	6	INCL	INCL
C3	Bitzer Semi-Hermetic Condensing Unit – Corridor area (+0°C evap) 15kw	No.	2	45,000	90,000
C4	Unit Coolers – Medium temp (Corridor) matching with above CU	No.	2	INCL	INCL
C5	Refrigerant piping, fittings, insulation, supports	Lot	Lot	INCL	INCL
C6	Refrigerant charge (R-404A)	Lot	Lot	INCL	INCL
	SUB-TOTAL C – REFRIGERATION				
D	ELECTRICAL WORKS				<u>660,000</u>
D1	Power distribution to all equipment	Lot	1	BY CLIENT	BY CLIENT
D2	Control panels with DIXELL ITALY controllers (3 rooms)+ 1 Corridor	No.	4	3,000	12,000
D3	LED cold-rated lighting (all rooms, IP65)	Lot	4	2,500	10,000
	SUB-TOTAL D – ELECTRICAL				<u>22,000</u>
E	INSTALLATION, TESTING & COMMISSIONING				
E1	Panel installation, sealing, and finishing	Lot	1	INCL	INCL

E2	Refrigeration system installation and piping	Lot	1	INCL	INCL
E3	Testing, commissioning, blast freezer pull-down verification	Lot	1	INCL	INCL
E4	Mobilization, project management	Lot	1	INCL	INCL
	SUB-TOTAL E – INSTALLATION & COMMISSIONING				
	GRAND TOTAL				1,213,750 QAR

Total Contract Value: QAR 1,213,750

7. COMMERCIAL TERMS & CONDITIONS

7.1 Payment Terms

TO BE DISCUSSED AND AGREED

7.2 Validity

This proposal is valid for thirty (30) calendar days from the date of submission.

7.3 Delivery & Installation Period

Estimated project duration is 4-12 weeks from contract signing and receipt of advance payment, subject to site readiness (completed civil works, slab, and building envelope).

9. WARRANTY

MES shall provide a comprehensive (1) year warranty from successful commissioning covering all supplied equipment, materials, and workmanship. Post-warranty AMC offered at competitive rates.

10. EXCLUSIONS & CLARIFICATIONS

Our scope does not include civil works, removal of existing units, steel base frames / foundation for condensers, machine room, closed working room and storage on site, electrical power supply on site during installation, plumbing and sanitary works, electrical sub-distribution panel in machinery room, electrical connection between sub-distribution panel and our control panels, insurance policy, governmental fees, any other related additional costs and any supplies and services not especially listed herein.

Without limitation, the following items are explicitly excluded:

1. All kinds of civil works including passage/sleeve for refrigerant pipes through the walls, etc.
2. Electrical power supply and earthing to our control panels.
3. Drain points as required.
4. Foundation or steel platforms for the condensers.
5. Adequate safe storage area for all equipments and materials.
6. Temporary water and electricity during installation.
7. Disposal and carting away of sundry items.
8. Offloading of materials at site.
9. Any third party approvals.
10. Factory visits.

11. Dosing system (water treatment) for the evaporative condensers.
12. Any other item/service not specifically mentioned herein.
13. Any special specifications other than the standard.
14. Any pipe ventilation works.
15. Concrete works or floor finishes (concrete topping over XPS is by others).
16. Any steel platform to hang the outdoor/indoor units.
17. Any civil works and plinths.
18. Any electrical works (power supply, cabling, distribution boards, earthing — all by Electrical Contractor).
19. Fire detection and firefighting systems within cold rooms.
20. Building Management System (BMS) integration.
21. Racking and shelving systems within cold rooms.
22. Floor heating system for frost-heave prevention.

METRI ENGINEERING SERVICES W.L.L.

